Automotive Ethernet Devices

2.5/5/10GBASE-T1 to 2.5/5/10BASE-T RJ45 electrical port wit RJ45 port module
Supports 10G multimode and single-mode fiber

MC10GM MEDIA CONVERTER USER MANUAL 2.5/5/10G MULTI-GIGABIT MEDIA CONVERTER

MC10GM

SUPPORT：17265076798

2.5/5/10G BASE-T1
MEDIA CONVERTER

MC10GM IS AUTOMOTIVE ETH MEDIA CONVERTER

FEATURES：

- Full-duplex conversion between 2.5/5/10G BASE-T1
- Master/Slave switch
- 2.5/5/10G speed
- DIP config
- Status LED
- Metal shell
- H-MTD Connector
- SFP+ Connector
- 5-17V Input

MC10GM Media Converter,

It is a switchable 10G/multi-speed automitive Ethernet media converter supporting 2.5G, 5G, and 10G. Rosenberger H-MTD interface, it converts the automitive Ethernet 2.5/5/10G BASE-T1 to industrial-grade 2.5/5/10G standard Ethernet, which can then be connected to computers. The product realizes full-duplex physical layer conversion between 2.5/5/10G Base-T1 and 2.5/5/10G Base-R, enabling direct point-to-point conversion between ECUs using 2.5/5/10G Base-T1 and any 2.5/5/10G standard Fast Ethernet devices. (No packets are stored or modified).

Architecture：

Pin out：

Pin	Funciton	Connector
1	2.5/5/10G BASE-T1 positive	Rosenberger H-MTD
2	2.5/5/10G BASE-T1 negative	Rosenberger H-MTD

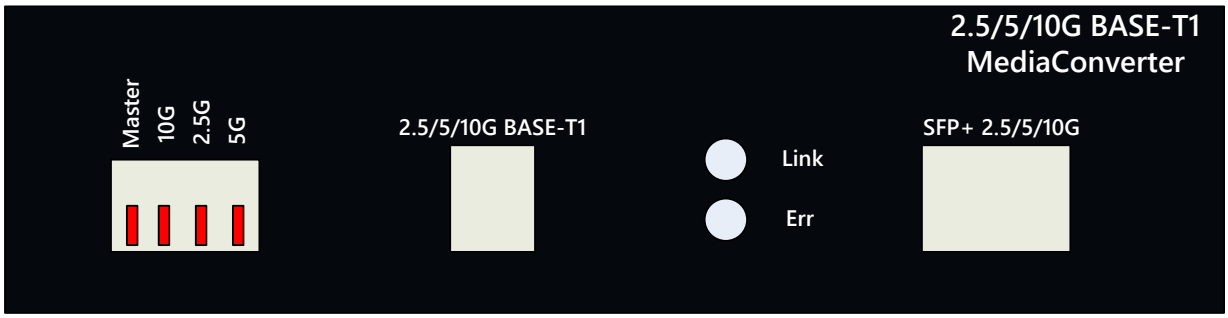
Parameters：

- 5V~17V input, compatible with passenger/commercial vehicles, electric vehicles, in-vehicle radar, etc.
- Temperature range: -40℃ to +85℃
- 2.5G/5G/10G speed, Master and Slave modes, DIP setting
- overvoltage protection, reverse polarity protection, and surge protection, suitable for long-term vehicle operation
- Dimensions: 112 x 67 x 32 mm

OPERATING

1DIP SETTING

The product uses DIP switches(1) to configure Master and Slave. The DIP switches(2-3-4) speed key is used for product SPEED configuration. The following is the layout of the device interface and the location of the switch configuration:



Master/Slave mode:
Set the 1st bit of the DIP switch to ON to configure it as Master mode.
Set the 1st bit of the DIP switch to OFF to configure it as Slave mode.

Speed select:
Set the 2nd bit ON and the 3rd and 4th bits to OFF to configure the device's operating speed to 10G BASE-T1
Set the 3nd bit ON and the 2rd and 4th bits to OFF to configure the device's operating speed to 2.5G BASE-T1
Set the 4nd bit ON and the 2rd and 3th bits to OFF to configure the device's operating speed to 5G BASE-T1

Finally, after powering on the device, the configuration will take effect immediately.

Configuration Examples:

Example 1: Set the 2nd bit of the DIP switch to ON, the 3rd and 4th bits to OFF (configuring the speed to 10G), and set the Master switch to ON. The Master mode will take effect immediately. The device will then operate in the Master mode for 10G BASE-T1 to 10G BASE-R conversion.

Example 2: Set the 2nd bit of the DIP switch to ON, the 3rd and 4th bits to OFF (configuring the speed to 10G), and set the Master switch to OFF. The Slave mode will take effect immediately. The device will then operate in the Slave mode for 10G BASE-T1 to 10G BASE-R conversion.

The configuration methods for 2.5G and 5G speeds are the same as above.

Note: When configuring the speed, ensure that only one of the DIP switch bits corresponding to 2.5G/5G/10G is set to ON, and the other two are set to OFF.

2 BRIEF INSTRUCTIONS FOR USE

Connect the TRX-N and TRX-P wires of the MC10GM Converter to the tested target(DUT), plug 12V power, and then use a regular network cable to connect the computer and Converter.

Supply 12V power to the DUT to ensure that both devices have successfully powered on and operated. At this point: (The green indicator light indicates an error state and remains off during normal operation.)

If the DUT is 10G BASE-T1 and Slave, the converter device should be configured as 10GBASE-T1 and Master, if all connection are correct, then the yellow light will be on, that indicating the data connection is normal and data communication can proceed. Flashing yellow light indicates data transmission on the 10G BASE-T1 network.

If the DUT is 10G BASE-T1 and Master, the converter device should be configured as10G BASE-T1 and Slave, if all connection are correct, then the yellow light will be on, that indicating the data connection is normal and data communication can proceed. Flashing yellow light indicates data transmission on the 1000BASE-T1 network.

The operations should be the same process if the DUT is 2.5/5G BASET-1 or Master.

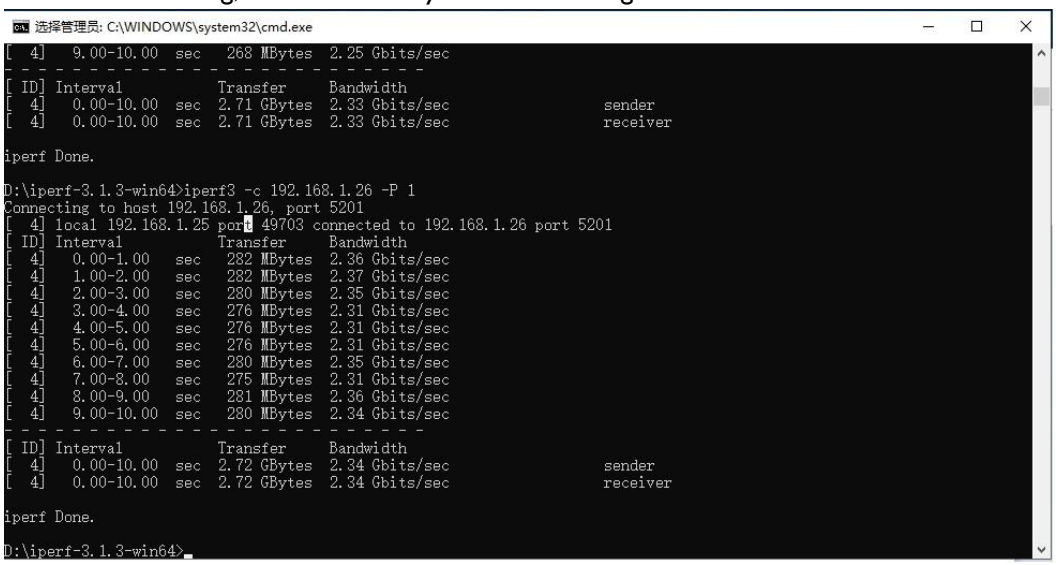
After the indicator light is normal, the Ethernet software on the computer can be used to monitor and collect data from the tested target.

Please ensure that the test computer and the test target are in the same network segment, and turn off the firewall function of the test computer.

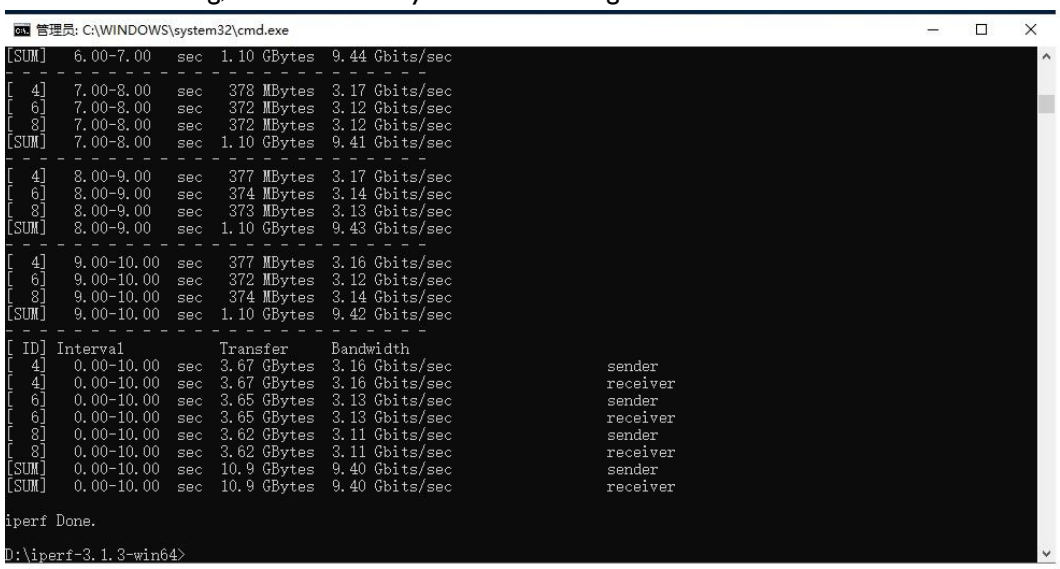
For example, the IP address of the tested target device is 192.168.1.122. Then configure the IP address of the computer to 192.168.1.155, which is in the same local area network segment. In this way, you can open the DOS terminal on your computer and use the “ping 192.168.1.122 – t” command(Excluding quotation marks) to send data packets to the target device.

You can also use the iperf3 software to perform bandwidth testing. The bandwidth test results using iperf3 are as follows:

In the 2.5G mode, due to limitations of the test computer and its operating system, the converter can achieve a rate of 2.34G at the 2.5G setting, which is basically close to the target value.



In the 10G mode, due to limitations of the test computer and its operating system, the converter can achieve a rate of 9.4G at the 10G setting, which is basically close to the target value.



Of course, you can also use the data collection software and Ethernet transceiver testing tools we provide:

Wireshark Tool:

https://wwm.lanzouh.com/iK3eQtjvfti Password:d5te

Data send tool:

https://wwm.lanzouh.com/i3bNTtjv7hi Password:4pd7

Testing tool:

https://wwm.lanzouo.com/b01psw5vi Password:bn3z